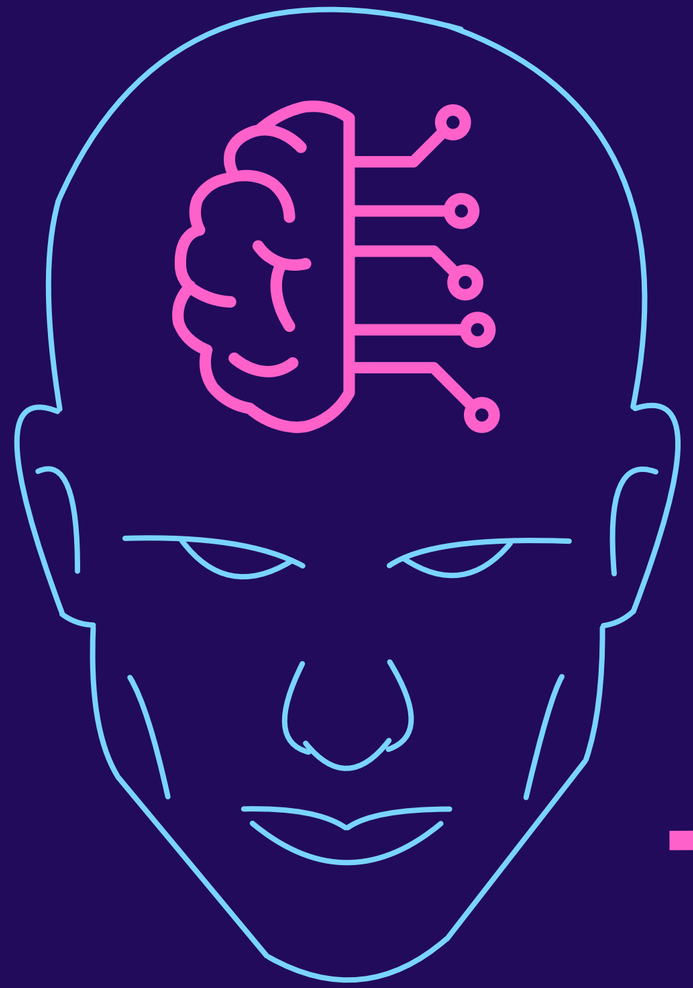
mitigating algorithmic bias



thought experiment

imagine an AI system that predicts future risk of lung cancer. it routinely underestimates the risk of cancer for black patients, incorrectly classifying high risk patients as low risk; yet routinely correctly estimates risk for white patients.


level up
level down
change nothing

level up

system designed to artificially inflate predictions for disadvantaged groups

└ increases rates of low risk black patients to be classified as high risk

level down

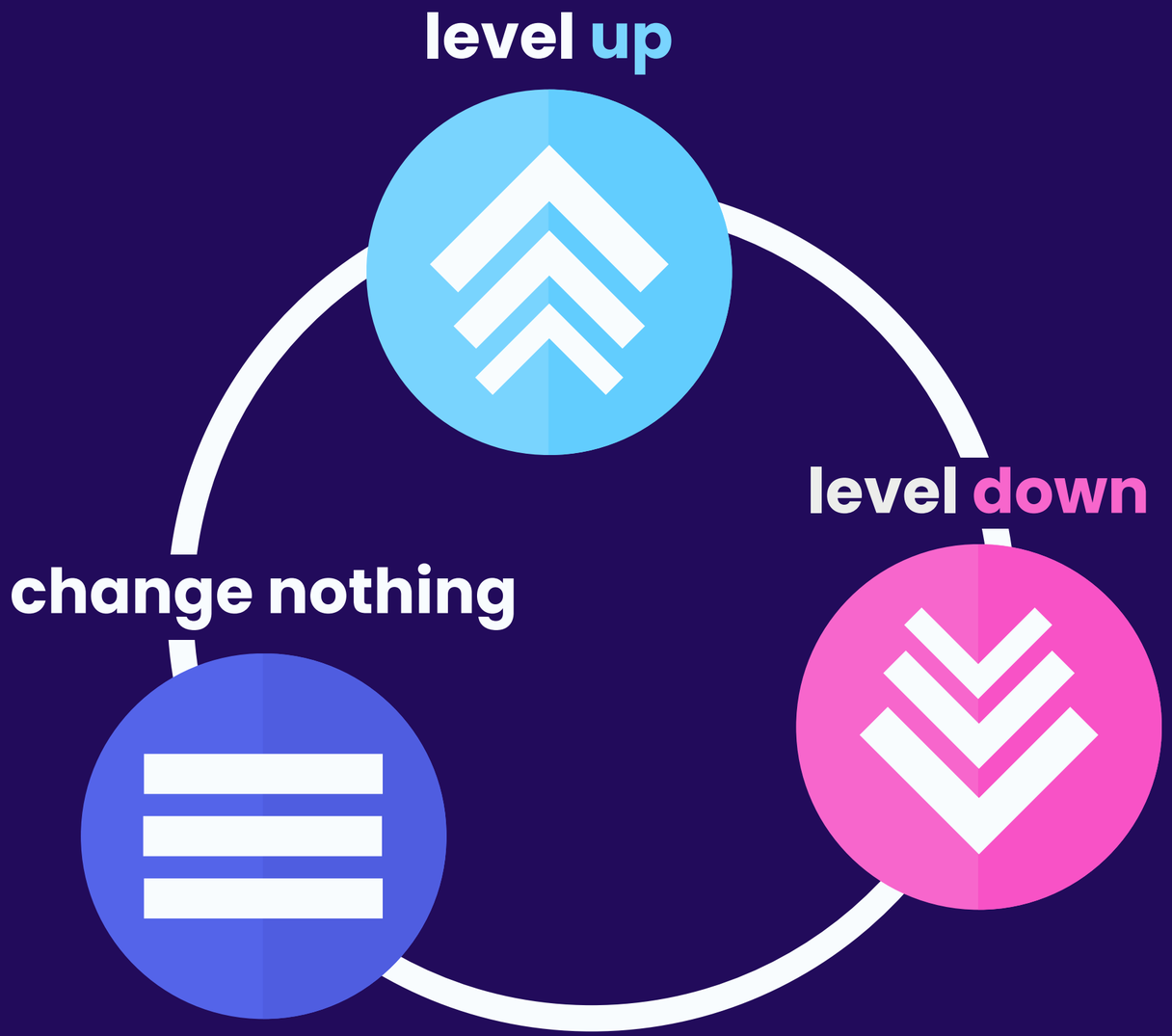
system decreases disparities by reducing advantages of privileged groups

└ adjust threshold for identifying high risk white patients so fewer are classified as high risk

change nothing

system maintains status quo and naturally corrects itself over time

└ the system would eventually sort out with specific measures


level up
level down
change nothing


level up
level down
change nothin

level up

arbitrarily makes a group better off to achieve balance between groups

└ risks greater number of overall false positives cancer patients

level down

less accurate for all groups, actively harmful for some groups

└ more overrepresented patients will be falsely negatively assessed

change nothing

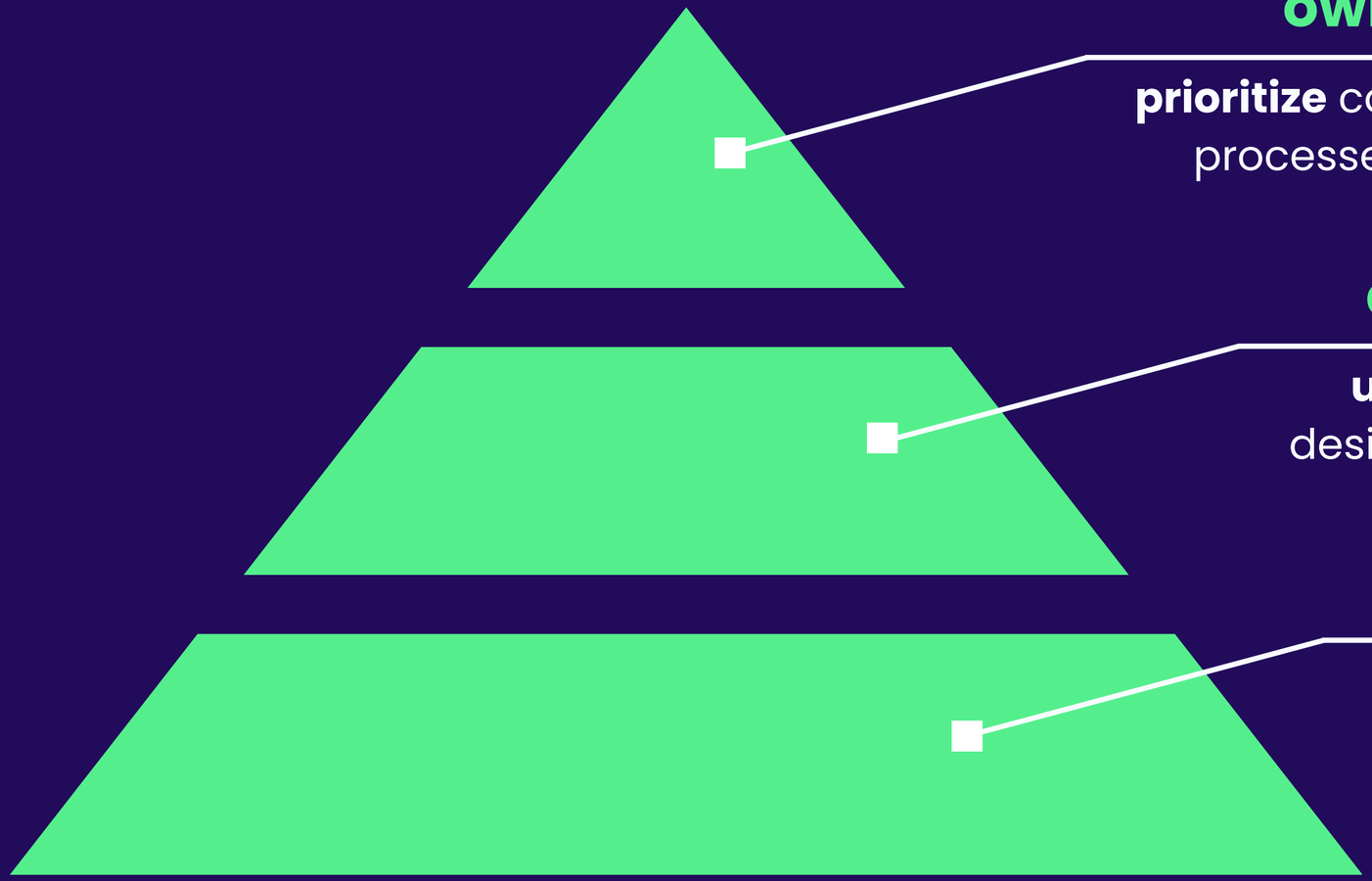
system continues to perpetuate disparities between groups

└ underrepresented patients will continue to be discriminated against

design justice

a multi-stakeholder participation approach

values of design justice



ownership

prioritize community design processes and outputs

accountability

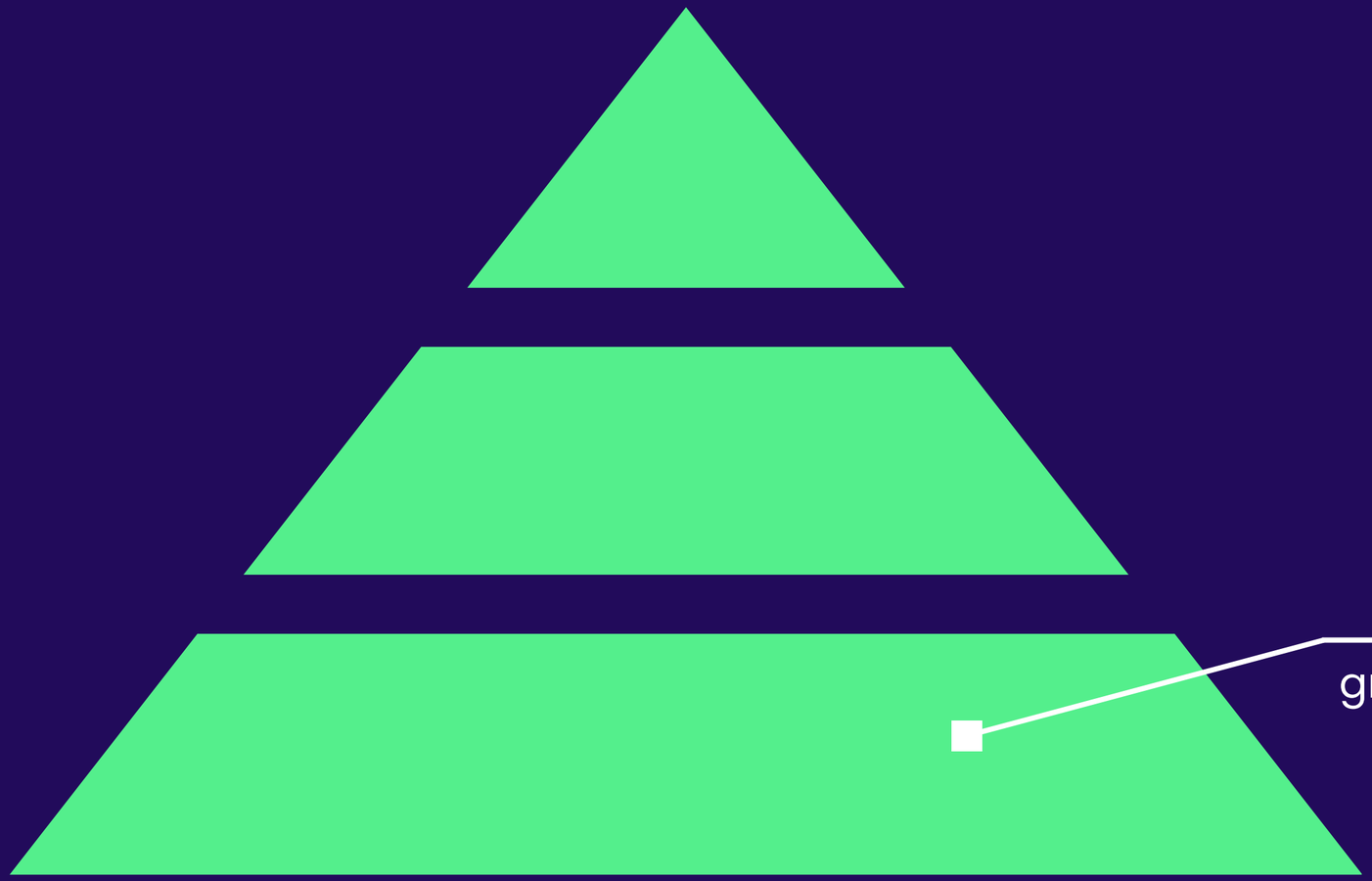
user-centered design
design driven by those most affected

equity

diverse workplace
diverse user base

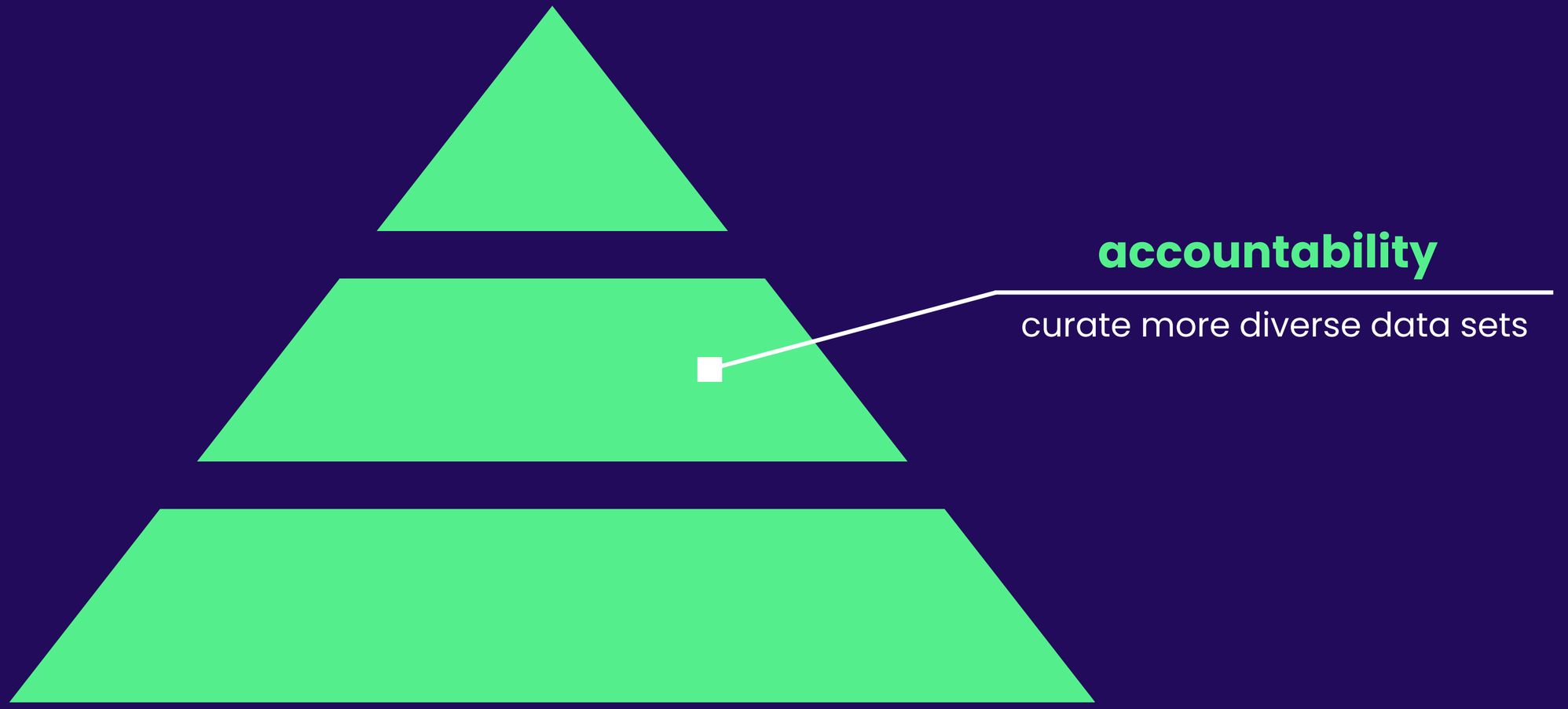
thought experiment

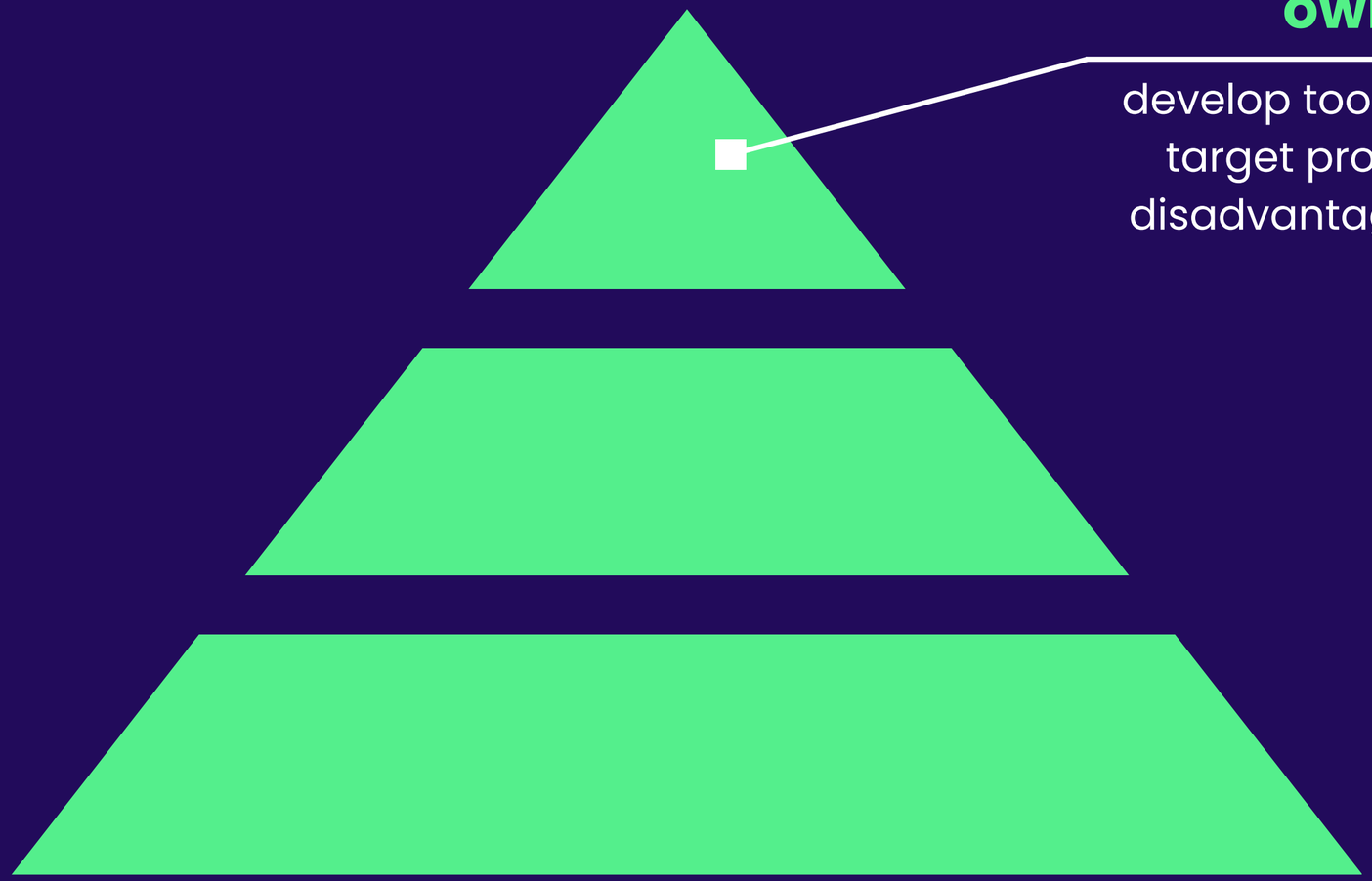
imagine an AI system that predicts future risk of lung cancer. it routinely underestimates the risk of cancer for black patients, incorrectly classifying high risk patients as low risk; yet routinely correctly estimates risk for white patients.



equity

greater access to health care & health care systems


accountability
curate more diverse data sets



ownership

develop tools that specifically target problems faced by disadvantaged communities

design justice

a multi-stakeholder participation approach

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